



On/Off Remote Control For Two Appliances

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This project will enable remote control of two electrical appliances. It makes use of a TV or DVD remote that uses modulated infrared (IR) pulse train operating at 38kHz frequency.

A good thing about this project is that it does not use any microcontroller

and is based on a CD4017B decade counter IC and TSOP1738 IR receiver only to control two appliances. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the two-channel IR remote control switch shown in Fig. 2

can be divided into two parts: regulated power supply and the control circuit.

The regulated power supply is built around a 12V-0-12V, 500mA step-down centre-tapped transformer (X1), full-wave rectifier using 1N4007 diodes (D1 and D2), 5V regulator IC 7805 (IC1), and two electrolytic capacitors C1 and C2.

The control circuit is built around IC CD4017B (IC2), 1N4148 diodes (D3, D4, D6, and D7), transistor BC557 (T1), two BC547 transistors (T2 and T3), and two 12V relays (RL1 and RL2).

The project requires 5V for CD4017B IC and TSOP1738 IR receiver (IRRX1) and 12V unregulated DC for the two 12V single-changeover SPDT relays.

The TSOP1738 IR receiver at input side receives 38kHz IR pulses from the remote control. Under normal condition, the output pin of receiver is at logic high, which means transistor T1 is cut-off and its collector terminal is at logic low. The collector of T1 drives clock line of the CD4017B decade counter.

When you point a remote towards IR receiver and press any key, the



Fig. 1: Author's prototype

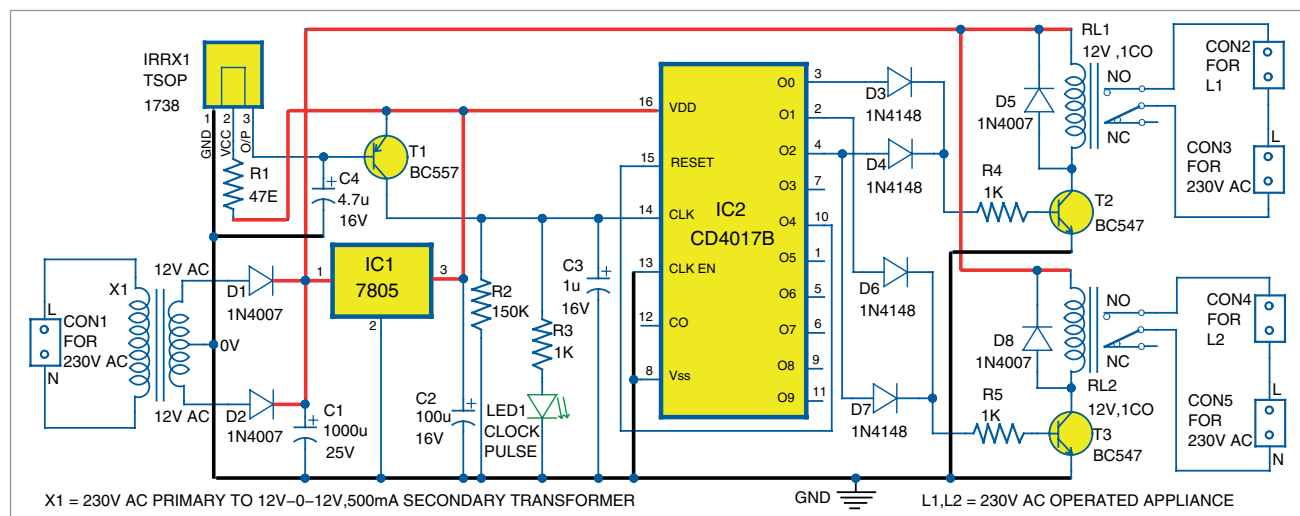


Fig. 2: Two-channel remote control circuit